KAIZEN MATHS

Lesson starter

Logarithms and Exponential Equations

Solving Equations

Name _____

S1

Solve: $4(2^x) = 16$

S2

Solve: $\log_2(x) = 2$

S3

Express using common logarithms: $\log_3(30)$

S4

Solve: $\ln(2x + 1) = \ln(x + 6)$

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Exit ticket

Logarithms and Exponential Equations

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E1

Solve: $5^x = 25$

E2

Solve: $\log_2(x) = 4$

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Express using common logarithms: $\log_3(25)$

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Teacher answers & working

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S1

Solve: $4(2^x) = 16$

Answer:

$x = 2$

Working:

Method focus: Use inverse operations to move between exponential and logarithmic form.

Identify the multiplier attached to the exponential term

The coefficient 4 is multiplying the whole exponential term 2^x.

Divide both sides by 4

$2^x = 16 / 4 = 4$

Rewrite as a matching power

$4 = 2^2$

Equate exponents

$x = 2$

Check: any logarithm argument must be positive, and exponential outputs must be positive.

Final answer: x = 2

S2

Solve: $\log_2(x) = 2$

Answer:

$x = 4$

Working:

Method focus: Use inverse operations to move between exponential and logarithmic form.

Convert to index form

$\log_2(x) = 2$ means $2^2 = x$

Calculate the power

$2^2 = 4$

$x = 4$

Check: any logarithm argument must be positive, and exponential outputs must be positive.

Final answer: x = 4

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S3

Express using common logarithms: $\log_3(30)$

Answer:

$\log(30) / \log(3)$

Working:

Method focus: Transform the equation into a simpler algebraic equation, then evaluate or check the result.

Recall the change of base formula

$\log_a(x) = \log_b(x) / \log_b(a)$

Choose the new base

Use base 10, written as log.

Substitute the values

$\log(30) / \log(3)$

Check: any logarithm argument must be positive, and exponential outputs must be positive.

Final answer: log(30) / log(3)

S4

Solve: $\ln(2x + 1) = \ln(x + 6)$

Answer:

$x = 5$

Working:

Method focus: Transform the equation into a simpler algebraic equation, then evaluate or check the result.

Use one-to-one property of ln

If ln(A) = ln(B), then A = B, provided both arguments are positive.

Set arguments equal

$2x + 1 = x + 6$

Solve

$x = 5$

$x = 5$

Check the domain

Both logarithm arguments are positive for this generated solution.

Check: any logarithm argument must be positive, and exponential outputs must be positive.

Final answer: x = 5

Teacher answers & working

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E1

Solve: $5^x = 25$

Answer:

$x = 2$

Working:

Method focus: Use inverse operations to move between exponential and logarithmic form.

Write the right side as a power of the same base

$25 = 5^2$

Equate exponents

$5^x = 5^2$ so $x = 2$

$x = 2$

Check: any logarithm argument must be positive, and exponential outputs must be positive.

Final answer: x = 2

E2

Solve: $\log_2(x) = 4$

Answer:

$x = 16$

Working:

Method focus: Use inverse operations to move between exponential and logarithmic form.

Convert to index form

$\log_2(x) = 4$ means $2^4 = x$

Calculate the power

$2^4 = 16$

$x = 16$

Check: any logarithm argument must be positive, and exponential outputs must be positive.

Final answer: x = 16

Teacher answers & working

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E3

Express using common logarithms: $\log_3(25)$

Answer:

$\log(25) / \log(3)$

Working:

Method focus: Transform the equation into a simpler algebraic equation, then evaluate or check the result.

Recall the change of base formula

$\log_a(x) = \log_b(x) / \log_b(a)$

Choose the new base

Use base 10, written as log.

Substitute the values

$\log(25) / \log(3)$

Check: any logarithm argument must be positive, and exponential outputs must be positive.

Final answer: log(25) / log(3)